

Assignment No: 04

Page No.	
Date	

① int n=4

for (int i=1; i<=n; i++) {

for (int j=1; j<=n; j++) {

sop(i);

}

}

x x x x
x x x x
x x x x
x x x x

time complexity = $n \times n = n^2 = O(n^2)$

space complexity = $O(1)$

② int n=4

for (int i=1; i<=n; i++) {

for (int j=1; j<=n; j++) {

sop(j);

}

}

1 2 3 4
1 2 3 4
1 2 3 4
1 2 3 4

time complexity = $n \times n = O(n^2)$

space complexity = $O(1)$

③ int n=4

for (int i=1; i<=n; i++) {

for (int j=1; j<=n; j++) {

sop(i);

}

}

1 1 1 1
2 2 2 2
3 3 3 3
4 4 4 4

time complexity = $n \times n = O(n^2)$

space complexity = $O(1)$

④

```

int n = 5;
for (int i = 1; i <= n; i++) {
    * * * * *
    for (int j = 1; j <= n; j++) {
        * * * * *
        if (i % 2 == 0 || j % 2 == 0) {
            * * * * *
            cout << " ";
        } else {
            * * * * *
            cout << "X";
        }
    }
    cout << endl;
}

```

Time complexity = $n \times n = O(n^2)$

Space complexity = $O(1)$

⑤

```

int n = 5;
for (int i = 1; i <= n; i++) {
    * * * * *
    for (int j = 1; j <= n; j++) {
        * * * * *
        if (i + j == n + 1 || i == j) {
            * * * * *
            cout << "X";
        } else {
            * * * * *
            cout << " ";
        }
    }
    cout << endl;
}

```

Time complexity = $n \times n = O(n^2)$

Space complexity = $O(1)$


```

⑥ For (int i=0; i<n; i++) {
    For (int j=1; j<=i; j++) {
        top(x);
    }
    top();
}

```

time complexity = $n \times n = O(n^2)$
 space complexity = $O(1)$

```

⑦ For (int i=5; i>=1; i--) {
    For (int j=1; j<=i; j++) {
        top("x");
    }
    top();
}

```

time complexity = $n \times n = O(n^2)$
 space complexity = $O(1)$

```

⑧ For (int i=0; i<n; i++) {
    For (int j=1; j<=n; j++) {
        if ((i+j)%2 == 0) {
            top(x);
        }
        else {
            top("H");
        }
    }
}

```

time complexity = $n \times n = O(n^2)$
 space complexity = $O(1)$


```

for (int j=1; j<=1; j++) {
    sop(j+" ")
}
sop();
}

```

time complexity = $n^2 + 2n^2 + n^2 + n^2 + 2n^2 + n^2 +$
 $= 8n^2$
 $= O(n^2)$

space complexity = $O(1)$

⑨

int n = 4

for (int i = 1; i <= n; i++) {

for (int j = 1; j <= n - i; j++) {

sop(" ");

}

for (int j = 1; j <= i; j++) {

sop(j);

}

}

time complexity = $n \times n + n \times n = 2n^2 = O(n^2)$ space complexity = $O(1)$

⑩

int n = 4

for (int i = 1; i <= n; i++) {

for (int j = 1; j <= i; j++) {

sop(j + " ");

}

for (int j = 1; j <= 2 * (n - i); j++) {

sop(" ");

}

for (int j = i; j >= 1; j--) {

sop(j + " ");

}

sop();

}

for (int i = n; i >= 1; i--) {

for (int j = 1; j <= i; j++) {

sop(j + " ");

}

for (int j = 1; j <= 2 * (n - i); j++) {

sop(" ");

}

n=3

```
for (int i=1; i<= 2*(n-1); i++) {  
    for (int j=1; j<= 2*(n-1); j++) {  
        int top = i-1;  
        int bottom = n 2*(n-1)-i;  
        int left = j-1;  
        int right = 2*(n-1)-j;  
  
        int layer = Math.min(Math.min(left, right),  
                               Math.min(top, bottom));  
  
        top (n-layer);  
    }  
    top();  
}
```

Time complexity = $O(n^2)$
Space complexity = $O(1)$

12

```
int n=5  
for (int i=0; i<=n; i++) {  
    int num = 1;  
    for (int j=1; j<=n-i; j++) {  
        top();  
    }  
    for (int j=0; j<=i; j++) {  
        top(num);  
        num = num * (i-j) / (j+1);  
    }  
    top();  
}
```

Time complexity = $O(n^2)$
Space complexity = $O(1)$


```

(13) int n = 4; int count = 1;
    for (int i = 1; i <= n; i++) {
        for (int j = 1; j <= n - i; j++) {
            op(" ");
        }
        for (int j = 1; j <= i; j++) {
            op(count);
            count++;
        }
        op("\n");
    }
    for (int i = n - 1; i >= 1; i--) {
        for (int j = 1; j <= n - i; j++) {
            op(" ");
        }
        for (int j = 1; j <= i; j++) {
            op(count);
            count++;
        }
        op("\n");
    }
}

```

```

      1
    2 3
  4 5 6
7 8 9 10
11 12 13
14 15
16

```

Time complexity = $n^2 + n^2 + n^2 + n^2$

$= 4n^2 = O(n^2)$

Space complexity = $O(1)$